Customer Transaction & Behavior Analysis

DSA1080A – Programming for Data Science

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1. Executive Summary

This report presents the findings of an end-to-end data analysis of a UK-based online retail dataset containing 541,909 raw transactions from December 2010 to December 2011. After cleaning, 392,692 transactions were retained across 4,338 unique customers and 37 countries, with 34 analytical columns derived from the original 8.

The analysis reveals that revenue is heavily concentrated — 81.97% originates from the United Kingdom, and a small high-value customer segment drives a disproportionate share of total spending. The repeat purchase rate of 65.6% indicates strong customer loyalty, while a pronounced Q4 seasonal pattern exposes the business to significant revenue fragility if peak-period demand is disrupted.

Three actionable recommendations are proposed: a high-value customer retention program, targeted international market development in the Netherlands and Australia, and an H1 revenue activation campaign to reduce seasonal dependency.

392,692 Cleaned Rows	4,338 Customers	£8.9M TOTAL REVENUE	65.6% Repeat Rate	81.97% UK Revenue Share

2. Introduction

2.1. Background

This project was completed as part of DSA1080A — Programming for Data Science at USIU-Africa. The objective was to apply a full data science workflow — from raw data ingestion through cleaning, exploratory analysis, and advanced insight generation — to a real-world dataset.

The domain selected is Customer Transaction & Behavior Analysis, chosen for its direct relevance to commercial data science roles and its alignment with the project's portfolio objectives.

2.2. Objectives

The primary objectives of this analysis are to:

- Understand the distribution of customer purchasing behavior
- Identify revenue drivers at the product, customer, and country level
- Detect seasonal and temporal purchasing patterns
- Segment customers by value and purchasing frequency
- Generate actionable business recommendations from data-driven findings

2.3. Research Questions

The following five research questions guided the analysis:

_RQ1: What is the distribution of purchase frequency across customers, and what proportion are high-frequency buyers?

_RQ2: How does revenue per customer vary, and which customer segment contributes most to total revenue?

_RQ3: What is the ratio of repeat to new customers, and what is the probability of a second purchase?

_RQ4: How is spending distributed across the customer base, and what does the distribution shape reveal?

_RQ5: Which countries generate the highest revenue, and how does purchasing behavior differ across regions?

3. Dataset Overview

Dataset Name	E-Commerce Data (Online Retail Dataset)
Source	UCI Machine Learning Repository via Kaggle
Link	https://www.kaggle.com/datasets/carrie1/ecommerce-data
Raw Rows	541,909
Cleaned Rows	392,692
Columns (after feature engineering)	34
Date Range	December 2010 — December 2011
Countries	37
Unique Customers	4,338
Unique Products	3,665

3.1. Variables Descriptions

Variable	Type	Description
InvoiceNo	Object	Unique transaction ID. Prefix 'C' indicates cancellation.
StockCode	Object	Unique product code.
Description	Object	Product name.
Quantity	Integer	Units purchased per transaction line.
InvoiceDate	Datetime	Date and time of transaction.
UnitPrice	Float	Price per unit in GBP (£).
CustomerID	Object	Unique customer identifier.
Country	Object	Customer's country of residence.
Revenue	Float	Derived: Quantity × UnitPrice.
Month	Object	Extracted from InvoiceDate (YYYY-MM format).
DayOfWeek	Object	Day name extracted from InvoiceDate.
Hour	Integer	Hour extracted from InvoiceDate.

Additional derived columns include Year, Quarter, MonthNum, WeekOfYear, DayOfMonth, RevenuePerUnit, PriceCategory, QuantityCategory, IsHighValue, CustomerOrderCount, CustomerTotalRevenue, CustomerAvgOrderValue, ProductTotalRevenue, ProductTotalQuantitySold, ProductOrderCount, CountryTotalRevenue, IsUK, InvoiceItemCount, InvoiceTotalRevenue, and InvoiceTotalQuantity.

4. Methodology

4.1. Tools and Environment

Tool	Purpose
Python 3	Primary programming language
Jupyter Notebook	Development and submission environment
pandas	Data loading, cleaning, and manipulation
matplotlib & seaborn	Data visualization
numpy	Numerical operations
GitHub	Version control and project submission

4.2. Statistical Methods Applied

_**IQR-based outlier detection** — used to contextualize zero-price and negative quantity removal decisions

_**Mean vs median comparison** — applied to detect distribution skewness across revenue and spending columns

_**Percentile-based segmentation** — used to segment customers into Low, Mid, and High spenders after standard deviation segmentation was evaluated and rejected due to extreme right skew producing a negative lower bound

_**Correlation analysis** — Pearson correlation applied to measure the relationship between purchase frequency and total customer spend

_**Probability estimation** — repeat purchase probability derived from the proportion of customers placing more than one order

_**RFM scoring** — quartile-based Recency, Frequency, and Monetary scoring applied to identify the most commercially valuable customers

Full implementation code is available in the project notebook at the GitHub repository:

https://github.com/Gad-rb/DSA1080A-SS2026-GadRubuye-CustomerTransactionAnalysis.git

5. Data Cleaning & Preparation

The raw dataset required significant preparation before analysis could begin. The following cleaning decisions were applied sequentially:

_**Missing Values:** CustomerID was missing in 24.9% of rows. Since all customer-level analysis depends on this identifier, affected rows were dropped. Description nulls (under 0.3% of rows) were also removed as product identity could not be resolved.

_**Duplicate Removal:** Fully duplicate rows were identified and removed to prevent double-counting of transactions.

_**Data Type Corrections:** InvoiceDate was converted from string to datetime to enable all time-based analysis. CustomerID was converted from float to string to prevent accidental arithmetic on an identifier field.

_**Cancellation Removal:** Transactions with InvoiceNo prefixed by 'C' were removed as they represent reversed purchases that would distort revenue totals and purchase frequency counts. Rows with negative Quantity or zero UnitPrice were also removed.

_**Revenue Derivation:** A Revenue column was created as Quantity × UnitPrice. This is the primary financial metric underpinning all downstream analysis.

_**Feature Engineering:** 26 additional columns were derived from existing data — including date features (Year, Quarter, Month, DayOfWeek, Hour), financial features (RevenuePerUnit, PriceCategory, QuantityCategory), and aggregated customer, product, invoice, and country-level metrics.

Stage	Row Count
Row Dataset	541,909
After dropping missing CustomerID	406,829
After removing duplicates	401,604
After removing cancellations & Invalid Rows	392,692
Final cleaned dataset	**392,692**

6. Exploratory Data Analysis

6.1. Summary Statistics

Descriptive statistics across key numeric columns reveal consistent right skew. The mean revenue per transaction (£22.63) exceeds the median (£12.45), and customer total revenue shows an even more pronounced divergence — mean of £11,565 against a median of £2,662. In skewed distributions, the median is the more representative measure of typical behavior.

6.2. Top 10 Products by Revenue

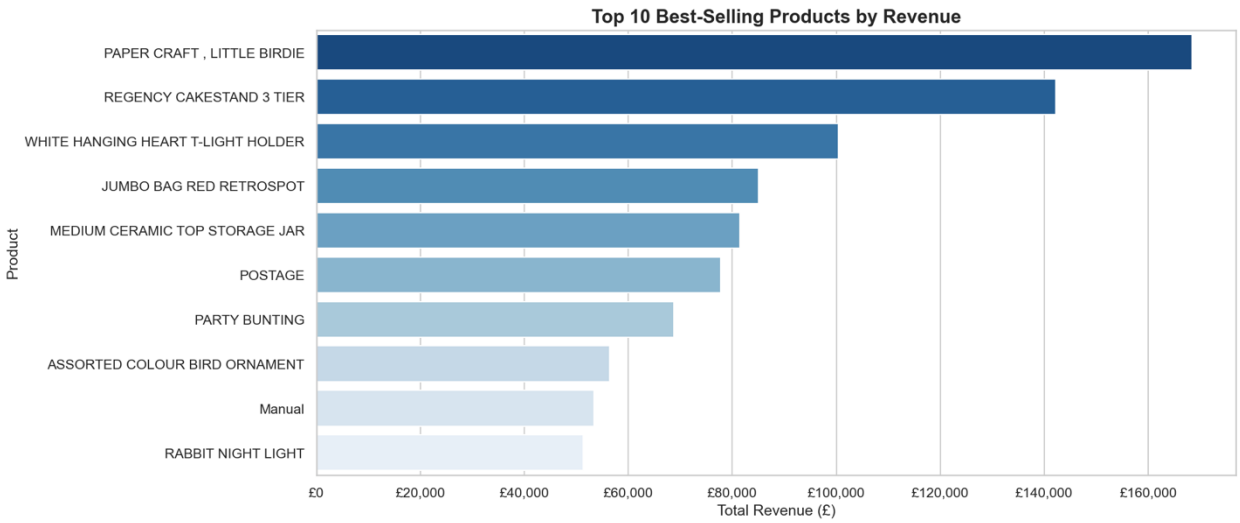

Figure 1: Top 10 Best-Selling Products by Revenue.

Revenue is concentrated among a small set of home décor, kitchenware, and gift products. Consistent with the retailer's wholesale catalogue profile. PAPER CRAFT, LITTLE BIRDIE leads all products at approximately £165,000, followed closely by REGENCY CAKESTAND 3 TIER at £140,000 and WHITE HANGING HEART T-LIGHT HOLDER at £100,000.

Notably, POSTAGE and Manual appear in the top 10. These are non-product entries that inflate the ranking and should be excluded from product-level recommendations.

The steep revenue drop-off from the third to the tenth product is consistent with the Pareto principle. It confirms that a core set of three to four items drives the majority of product revenue, warranting prioritized inventory management and targeted promotional support for these lines.

6.3. Monthly Revenue Trend

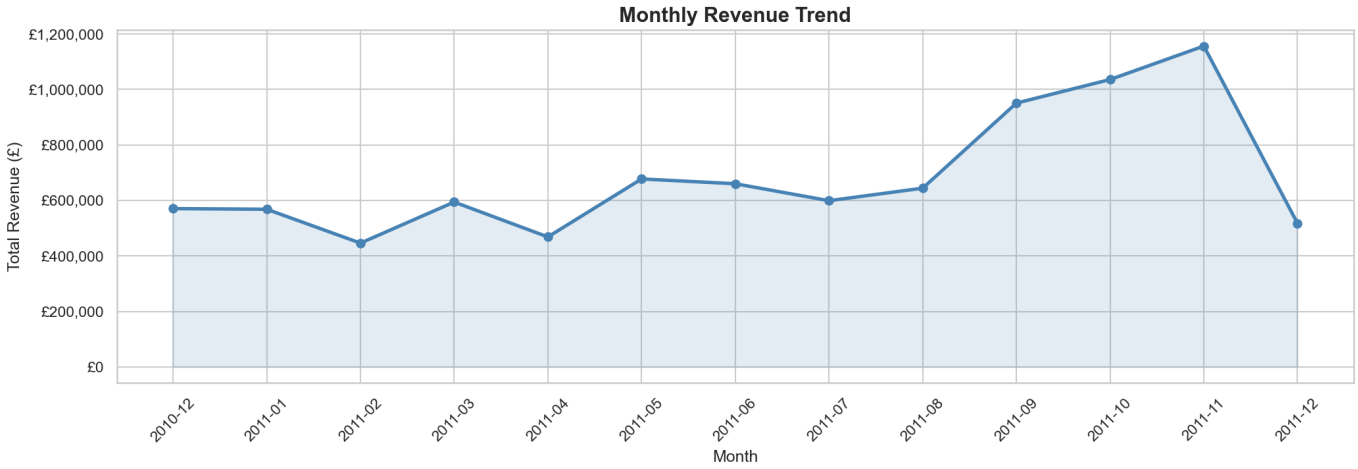

Figure 2: Monthly Revenue Trend from December 2010 to December 2011.

Revenue follows a clear seasonal pattern with a strong acceleration in Q4. The first half of the year maintains relatively stable revenue between £450,000 and £680,000, with February and April representing the two weakest months at £446,085 and £468,374 respectively.

From September onwards, revenue accelerates sharply by rising from £950,690 in September to a peak of £1,156,206 in November, driven by pre-holiday wholesale purchasing. December drops to £517,190, which reflects incomplete data for that month rather than an actual revenue decline.

This seasonal concentration in Q4 has direct implications for inventory planning, staffing, and promotional timing.

6.4. Customer Spending Distribution

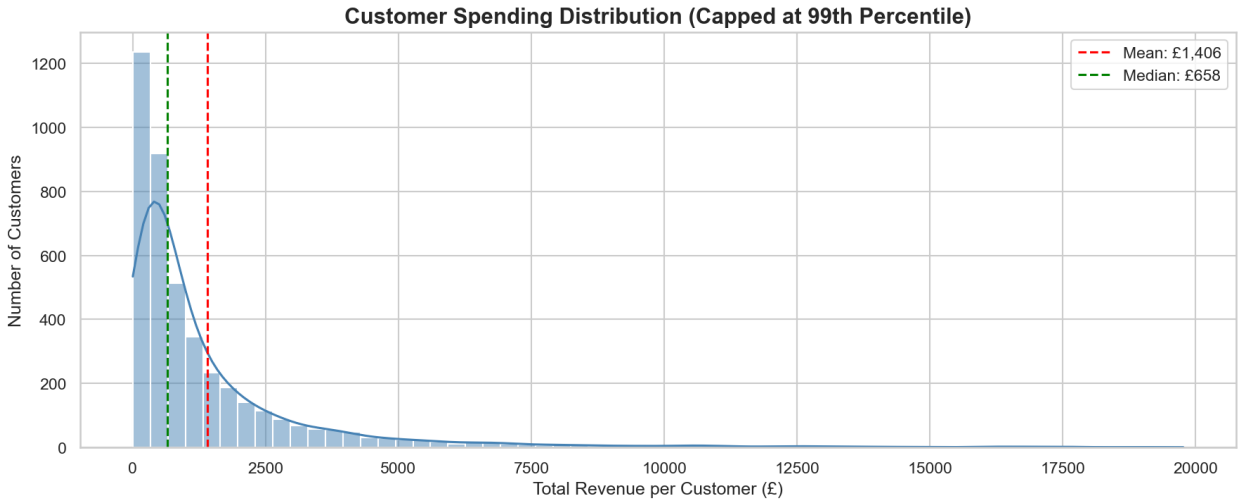

Figure 3: Customer Spending Distribution capped at 99th Percentile.

Customer spending is strongly right-skewed, as confirmed by the significant gap between the median (£658) and the mean (£1,406) within the 99th percentile cap. The majority of customers (over 1,200) cluster in the lowest spending bracket below £500, while the distribution tapers off sharply toward higher spending levels.

The mean being more than double the median is the statistical signature of right skew, confirming that a small number of high-spending customers pull the average upward disproportionately.

Even after capping at the 99th percentile, spending extends beyond £20,000 for a handful of customers, further illustrating the concentration of revenue in a narrow high-value segment.

The median of £658 is the more representative measure of typical customer spending behavior.

6.5. Revenue Distribution by Top 10 Country

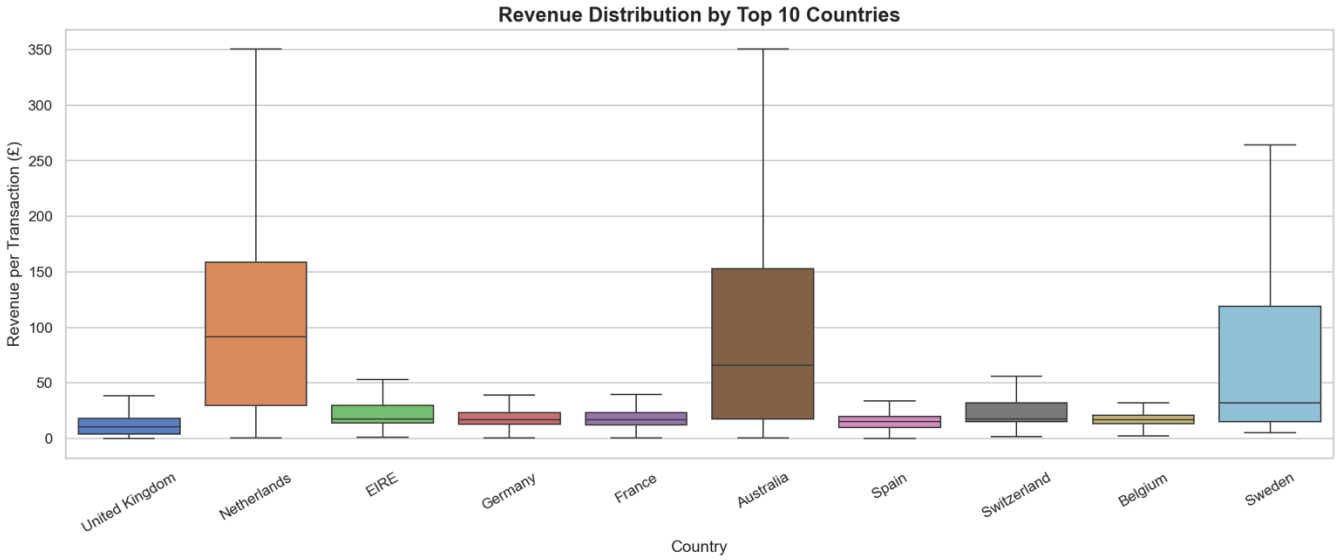

Figure 4: Revenue per transaction distribution across the top 10 countries by total revenue.

While the United Kingdom dominates total revenue at £7,285,024, its median transaction value is the lowest among all top 10 countries by sitting below £20 per transaction. This reflects the sheer volume of small, frequent domestic orders rather than high individual transaction values.

In contrast, the Netherlands and Australia stand out with significantly higher median transaction values of approximately £90 and £65 respectively and wide interquartile ranges extending to £350, indicating that these markets place fewer but substantially larger orders. Sweden also shows a notably elevated median and wide spread.

This pattern suggests that international customers, particularly from the Netherlands and Australia, behave as high-volume wholesale buyers placing large consolidated orders, while UK customers transact more frequently in smaller quantities.

These differences have direct implications for pricing strategy and account management by market.

6.6. ***Purchase Activity Heatmap***

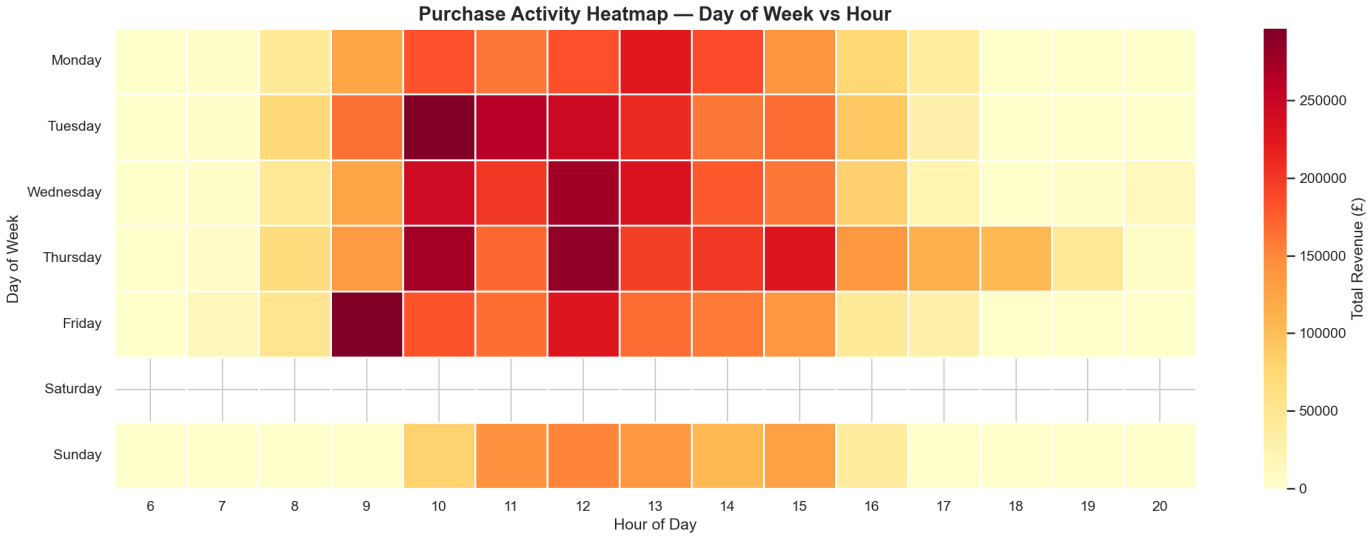

Figure 5: Purchase activity heatmap — revenue by day of week and hour of day.

Purchase activity is firmly concentrated within weekday business hours, with peak revenue generated between 09:00 and 15:00 from Tuesday to Thursday. The darkest cells, which are representing revenue above £250,000, cluster around 09:00–13:00 on Tuesday, Wednesday, and Thursday, identifying these as the highest-value purchasing windows in the week.

Friday shows an unusual early peak at 09:00, suggesting customers front-load their orders before the weekend. Saturday shows near-zero activity across all hours, and Sunday, while showing some mid-day activity between 10:00 and 15:00, remains significantly below weekday levels.

Thursday stands out as the only day with meaningful evening activity extending to 19:00–20:00. This weekday concentration strongly confirms a B2B wholesale customer base operating on standard business schedules, with Tuesday to Thursday mornings representing the optimal window for time-sensitive promotions and outreach campaigns.

7. Advanced Analysis

7.1. Customer Segmentation

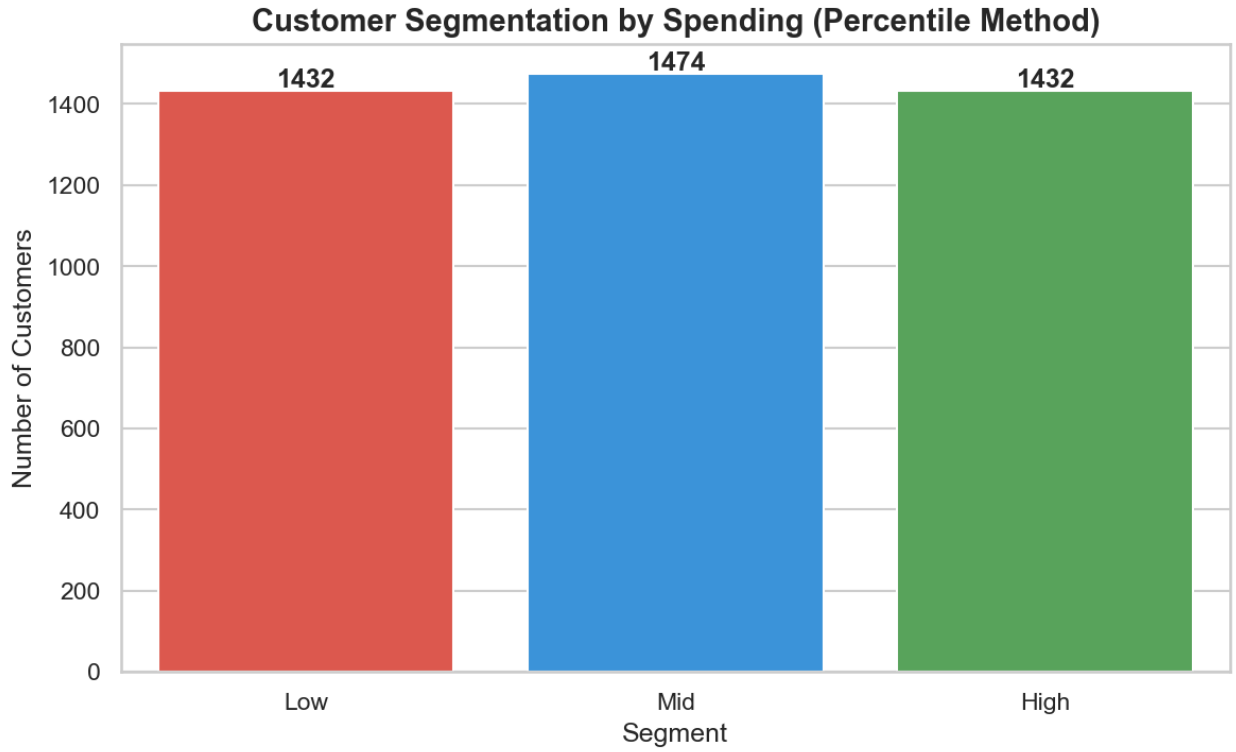

Figure 6: Customer segmentation using percentile-based thresholds.

Percentile-based segmentation divides the 4,338 customers into three balanced groups. Low spenders (below £384) account for 1,432 customers, Mid spenders (£384–£1,213) account for 1,474, and High spenders (above £1,213) account for 1,432.

The near-equal distribution across segments confirms that percentile thresholds are the appropriate method for this dataset. Standard deviation segmentation was evaluated but rejected because the extreme right skew produced a negative lower bound of £-6,937, which is statistically meaningless for spending data.

The High segment, representing 33% of customers, generates a disproportionate share of total revenue and warrants priority retention investment.

7.2. Repeat vs New Customers

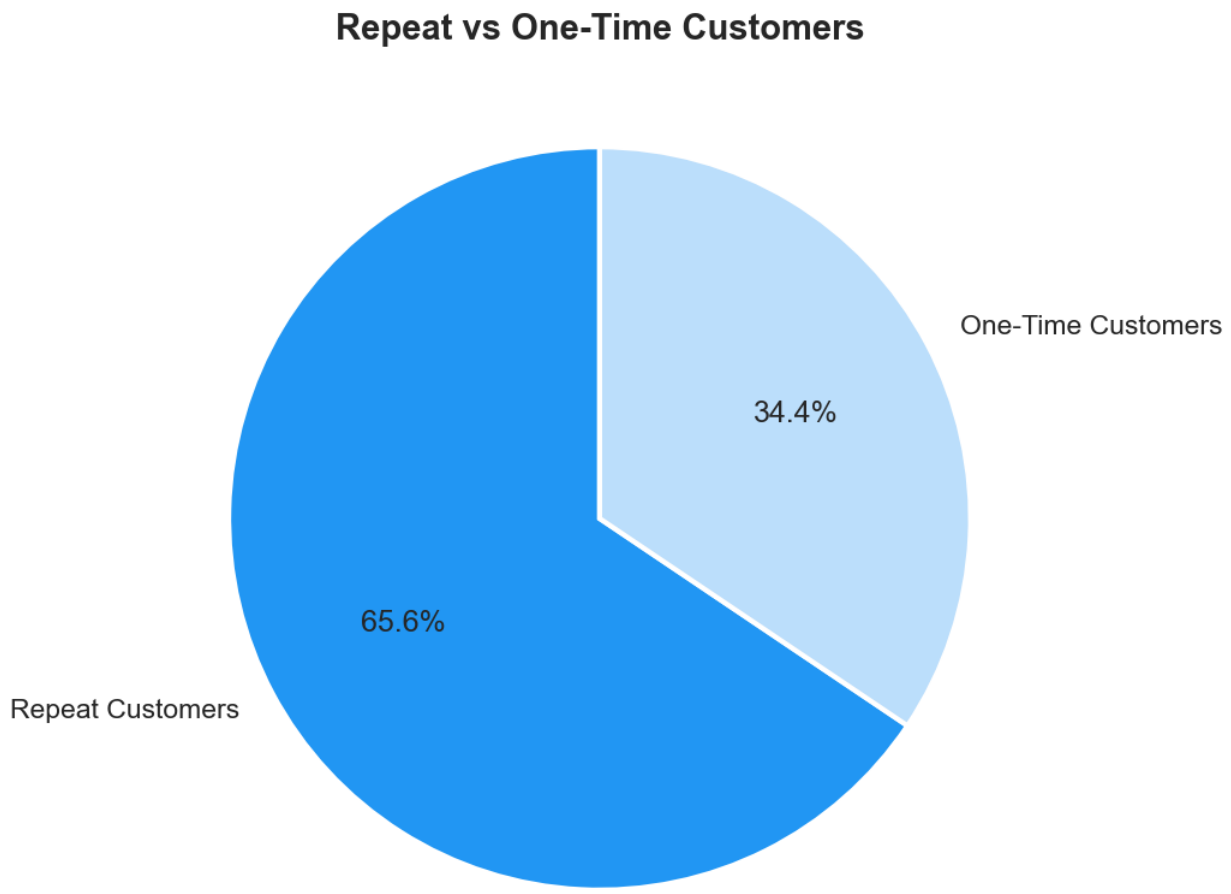

Figure 7: Repeat vs one-time customer breakdown.

65.6% of customers placed more than one order, with 2,845 repeat buyers against 1,493 one-time customers. A repeat purchase rate above 65% reflects strong customer loyalty for a wholesale retailer where long-term account relationships drive sustained revenue.

The average customer placed 4.27 orders, with the most active customer placing 209 orders which confirms that a core group of highly engaged buyers anchors the business.

The 1,493 one-time customers represent an untapped retention opportunity, converting even 20% of them to repeat buyers would add approximately 299 additional loyal accounts.

7.3. RFM Analysis

RFM scoring provides a multidimensional view of customer value that total revenue alone cannot capture. A customer who spent heavily a year ago but has not returned is less valuable than one who spent moderately last week. The top-scoring customers by RFM total are those who purchased recently, frequently, and at high monetary value. These represent the retailer's most engaged and commercially critical accounts. Customers with high Monetary but low Recency scores are lapsed high-value customers and represent the highest-priority re-engagement target, as they have demonstrated willingness to spend but have not returned.

7.4. Country Revenue Share

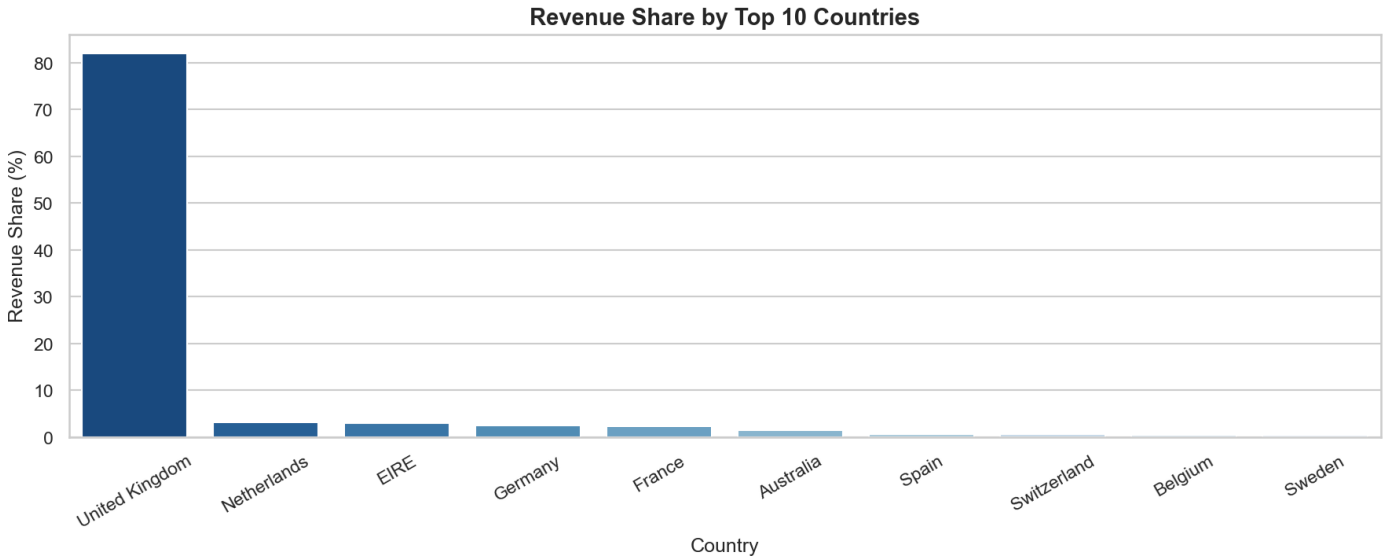

Figure 8: Revenue share by top 10 countries.

The United Kingdom accounts for 81.97% of total revenue, confirming extreme market concentration in the retailer's domestic market. The Netherlands (3.21%), Ireland (2.98%), and Germany (2.57%) are the next strongest markets but each hold less than 4% revenue share individually.

The remaining six countries in the top 10 (France, Australia, Spain, Switzerland, Belgium, and Sweden) collectively account for less than 7% of total revenue.

This level of geographic concentration presents a structural business risk, as any disruption to UK operations would have an immediate and severe impact on overall revenue performance.

7.5. Peak Purchasing Periods

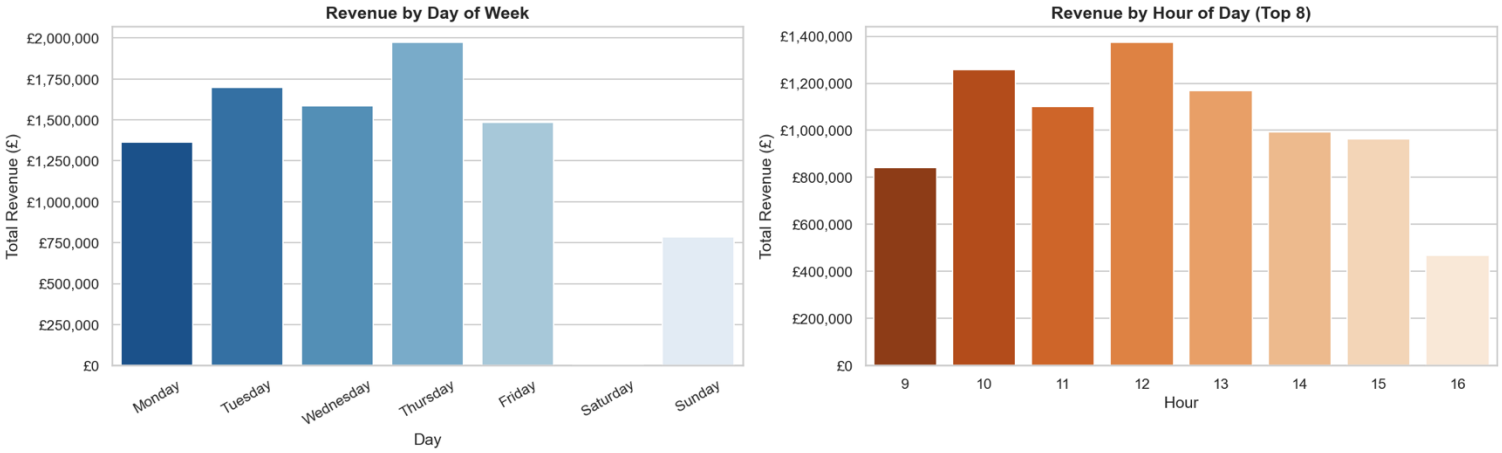

Figure 9: Revenue by day of week (left) and hour of day (right).

Thursday is the highest-revenue day at £1,973,016, followed by Tuesday at £1,697,734 and Wednesday at £1,584,284. Saturday recorded zero transactions, confirmed by a NaN value in the aggregation and Sunday generated only £785,490, the lowest of any active day.

This sharp weekday concentration is definitive evidence of a B2B wholesale customer base operating on standard business schedules.

The peak purchasing hour is 12:00 at £1,373,695, with strong activity sustained from 10:00 through 14:00.

Thursday midday therefore represents the single highest-value purchasing window in the week and the optimal moment for time-sensitive promotions and outreach.

7.6. Correlation: Purchase Frequency vs Total Spend

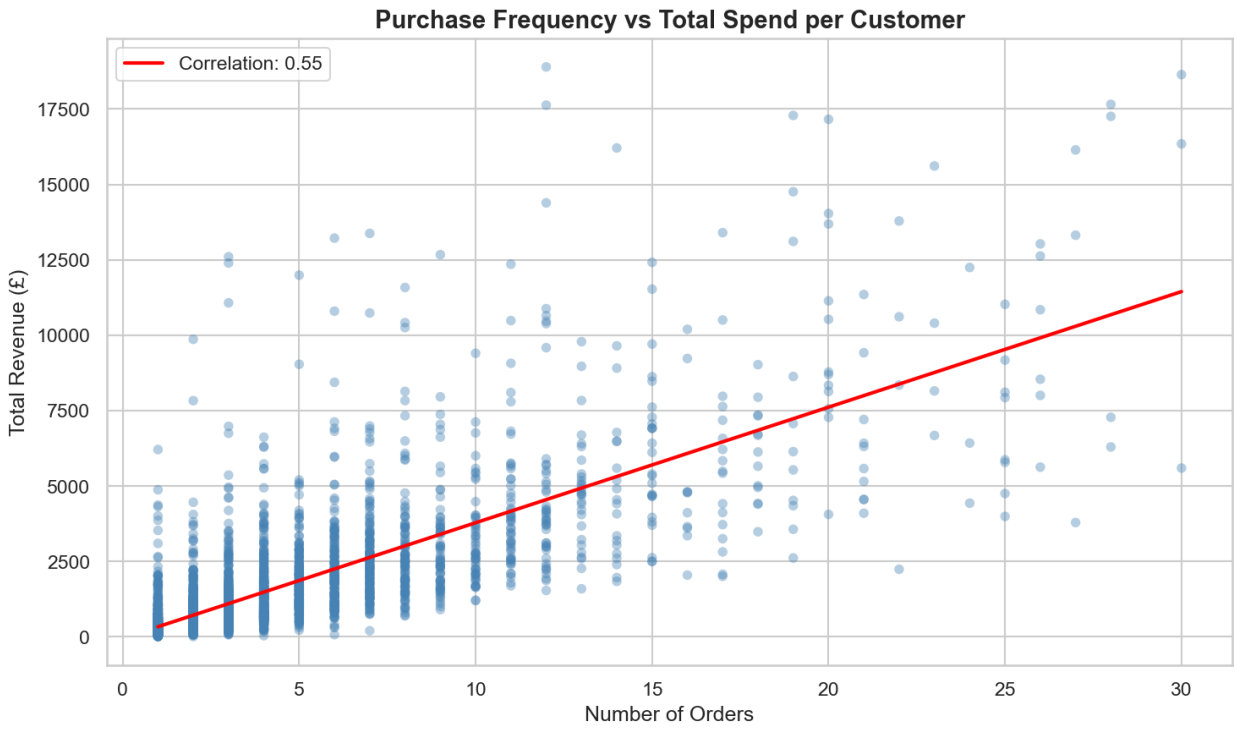

Figure 10: Scatter plot of purchase frequency vs total customer spend (capped at 99th percentile).

Purchase frequency and total spend show a moderate positive correlation of 0.55. This confirms a meaningful relationship; customers who order more frequently tend to spend more in total but the correlation is not perfect, indicating that order size also plays a significant independent role in determining customer value.

A customer placing fewer large orders can outspend a high-frequency buyer placing small ones.

This finding suggests that loyalty programs should reward both frequency and order value rather than frequency alone, to capture the full spectrum of high-value customer behavior.

8. Research Questions & Answers

RQ1 — Purchase frequency distribution: The average customer placed 4.27 orders, with a median of 2 orders. The distribution is strongly right-skewed — the top customer placed 209 orders while 34.4% of customers placed only one. High-frequency buyers (top quartile — 5+ orders) represent a small but commercially critical segment.

RQ2 — Revenue per customer and top segment: Customer total revenue ranges from £3.75 to £280,206 with a median of £668.57. The High spending segment — customers spending above £1,213 — accounts for 33% of customers but generates a disproportionate share of total revenue, confirming that this segment warrants priority retention investment.

RQ3 — Repeat vs new customers and repeat probability: 65.6% of customers placed more than one order. The probability of a customer making a second purchase is 0.656 — significantly above what would be expected in a transactional consumer retail context, reflecting the wholesale nature of the business and the strength of existing customer relationships.

RQ4 — Spending distribution shape: Customer spending is strongly right-skewed. The median spend of £658 sits well below the mean of £1,406, confirming that a small number of high-spending customers pull the average upward. The majority of customers cluster below £500 in total lifetime spend.

RQ5 — Country revenue and behavioral differences: The United Kingdom dominates at 81.97% of total revenue but records the lowest median transaction value among the top 10 countries. International markets — particularly the Netherlands and Australia — place fewer but significantly larger orders, suggesting distinct wholesale buying patterns that differ fundamentally from the domestic customer base.

9. Key Insights

9.1. Revenue is critically concentrated in a single market

The United Kingdom accounts for 81.97% of total revenue. The next nine markets combined contribute less than 18%. This level of geographic concentration creates structural business risk. Any disruption to UK operations would have an immediate and severe impact on overall performance.

9.2. A small high-value customer segment anchors total revenue

Percentile-based segmentation reveals that the top 33% of customers (those spending above £1,213) generate a disproportionate share of total revenue. The most active customer placed 209 orders and the highest spender generated £280,206. Retaining this segment is the single highest commercial priority.

9.3. Repeat purchase rate of 65.6% confirms strong customer loyalty

2,845 of 4,338 customers returned for a second order, with an average of 4.27 orders per customer. This retention rate is strong for a wholesale retailer. However, 1,493 one-time customers represent an untapped opportunity; converting 20% of them to repeat buyers would add approximately 299 loyal accounts.

9.4. Q4 seasonal concentration creates revenue fragility

November is the single highest revenue month at £1,156,205 (more than double February's £446,085). Over 40% of annual revenue is generated between September and November. Any Q4 disruption to supply chain, logistics, or demand would collapse annual targets.

9.5. International markets are underleveraged relative to their order value

The Netherlands and Australia place far fewer orders than the UK but at significantly higher median transaction values. Approximately £90 and £65 per transaction respectively versus under £20 for the UK. These markets demonstrate high-value wholesale buying behavior and represent the strongest candidates for targeted international expansion.

10. Recommendations

10.1. Launch a high-value customer retention program

Identify customers in the top RFM quartile and assign dedicated account management. Offer volume-based pricing incentives, early access to new products, and quarterly business reviews. Given that the top spending segment drives a disproportionate share of revenue, losing even a handful of these accounts would have material financial impact.

10.2. Invest in international market development for the Netherlands and Australia

Both markets demonstrate high median transaction values and wholesale buying patterns consistent with the retailer's core business model. A targeted expansion strategy, including localized account managers, country-specific promotions, and logistics partnerships, could significantly grow revenue from these markets without requiring proportional increases in transaction volume.

10.3. Implement an H1 revenue activation campaign to reduce Q4 dependency

February and April are the two weakest months at £446,085 and £468,374 respectively. A structured H1 promotional calendar, including early-year loyalty discounts, new product launches in January and February, and spring campaign bundles, would reduce seasonal revenue concentration, improve cash flow stability, and reduce the business risk created by over-reliance on Q4 performance.

11. Conclusion

This analysis of 392,692 cleaned transactions from a UK-based online retailer reveals a business with strong customer loyalty but significant structural concentration risks. Revenue is overwhelmingly dependent on a single geography, a small high-value customer segment, and a narrow Q4 seasonal window.

The analytical pipeline — spanning data cleaning, feature engineering, exploratory analysis, RFM scoring, customer segmentation, and correlation analysis — produced five key insights and three actionable recommendations grounded in statistical evidence rather than assumption.

The findings confirm that sustainable revenue growth requires three parallel priorities: retaining high-value customers through structured account management, activating underleveraged international markets, and smoothing seasonal revenue concentration through targeted H1 campaigns.

12. References

- UCI Machine Learning Repository — Online Retail Dataset: kaggle.com/datasets/carrie1/ecommerce-data
- McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
- Seaborn Documentation: seaborn.pydata.org
- Matplotlib Documentation: matplotlib.org
- pandas Documentation: pandas.pydata.org
- GitHub Repository: github.com/Gad-rb/DSA1080A-SS2026-GadRubuye-CustomerTransactionAnalysis

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